

ASSIGNMENT-2

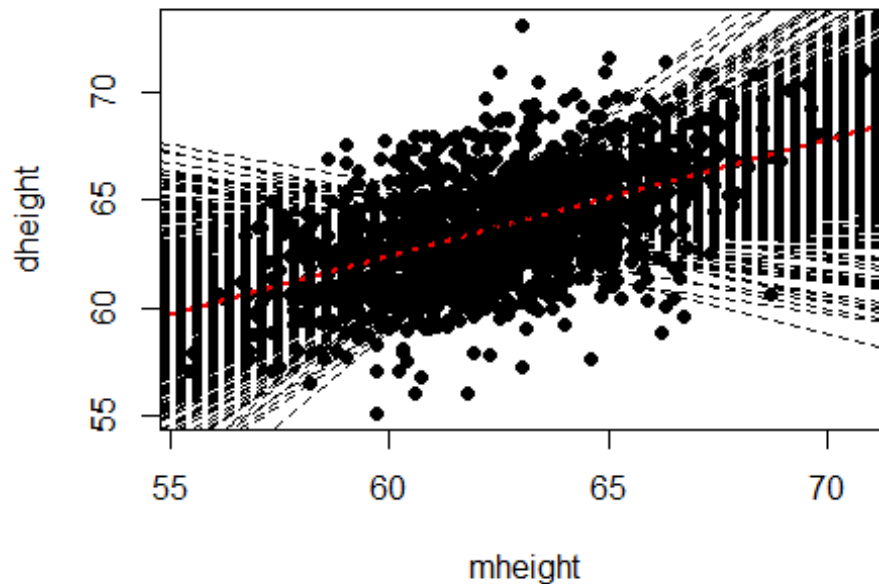
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2022-10-03

```
#####  
## ASSIGNMENT_2 #####  
#####  
  
library (alr4)  
  
#####  
  
#### Population + samples: coefficients  
  
the sample size I took here is 10  
size <- 10  
rep <- 500  
coefficients <- matrix (NA, ncol = 2, nrow = rep)  
for (i in 1: rep)  
{index <- sample (x = seq (1: dim (Heights) [1]), size = size, replace = FALSE)  
  
coefficients [i,] <- with (Heights [index,], lm (dheight ~ mheight) $ coef)}  
  
# Visualizing sampling distributions: many lines on the population plot  
with (Heights, plot (dheight ~ mheight, pch = 19, main = "Many lines for many samples"))  
with (Heights, abline (lm (dheight ~ mheight), lty = "solid", lwd = 2, col = "red"))  
for (i in 1: rep) {abline (coefficients [i,], lty = "dashed")}
```

For the sample size of 10, the band or the line that I had got is too extreme.

Many lines for many samples

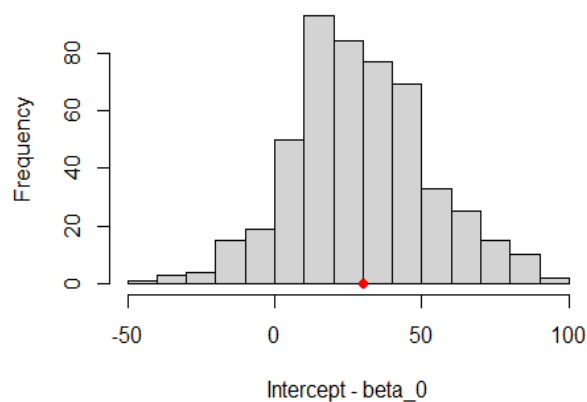


###SAMPLED DISTRIBUTION OF HAT(BETAS)#####

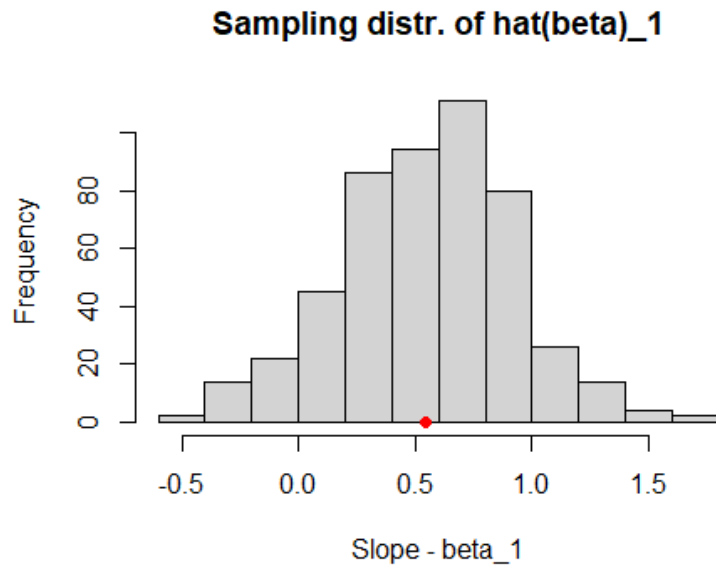
```
hist(coefficients[,1],xlab="Intercept - beta_0", main="Sampling distr. of hat(beta)_0")
with(Heights, points(0~lm(dheight~mheight)$coef[1],col="red",pch=19))
```

In the figures down I computed the frequency of $\hat{\beta}_0$ value and $\hat{\beta}_1$ for my sample distribution. The red dot in both figures represents the intercept and slope values of the main population distribution.

Sampling distr. of $\hat{\beta}_0$

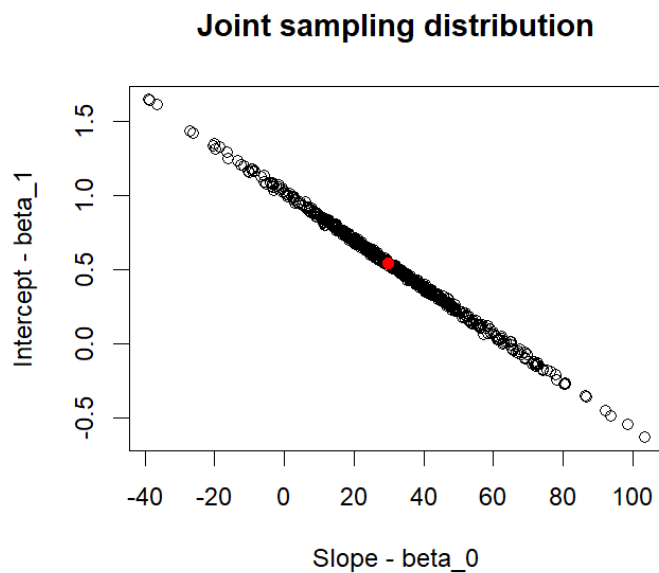


```
hist(coefficients[,2],xlab="Slope - beta_1", main="Sampling distr. of hat(b
eta)_1")
with(Heights, points(0~lm(dheight~mheight)$coef[2],col="red",pch=19))
```



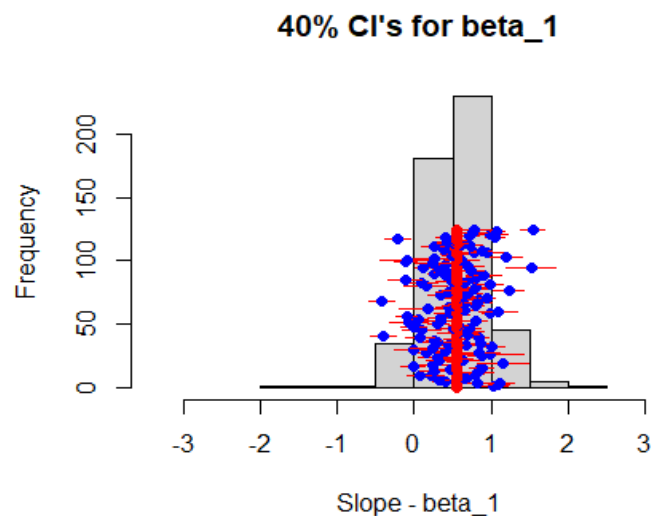
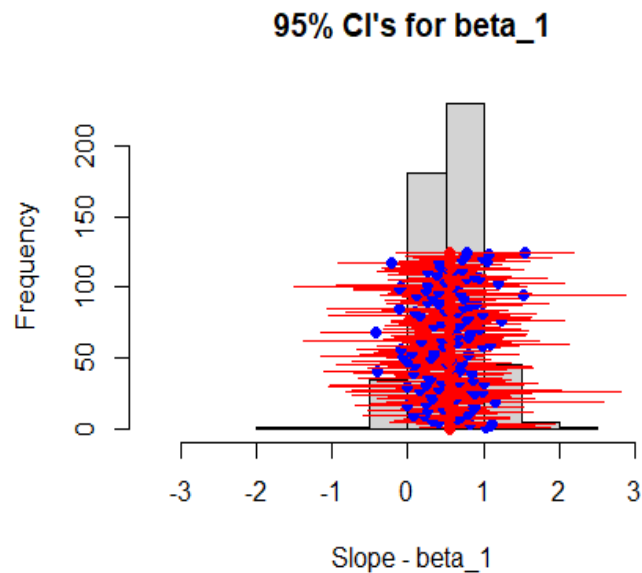
Joint sampling distribution of both hat(BETAS)##

```
plot(coefficients[,2]~coefficients[,1],xlab="Slope - beta_1", ylab="Intercept - beta_0", main=
"Joint sampling distribution")
with(Heights, points(lm(dheight~mheight)$coef[2]~lm(dheight~mheight)$coef[1],pch=19,col="red")
)
```



In the figure, I joined the two coefficients of my population distribution which I got from computing sample distribution. After the analysis, the intercept of the value is 29.917437, and the slope value is 0.541747.

##COMPARING CONFIDENCE INTERVALS OF HAT(BETA1) IN 95% and 40% ###



Here, I'm comparing the confidence intervals of slope in 95% and 40% . as you can see that the intervals that are distributed when the CI is 95% are more accurate and values are mostly near to the population slope value when compared to 40% of CI also the lengths of the intervals is small here.

```
#####
# PREDICTION
#####
```

```
with(Heights, plot(dheight~mheight))
with(Heights, abline(lm(dheight~mheight), lwd=2))
```

```
m0 <- lm(dheight~mheight,data=Heights)
predict(m0)
```

These are the predicted values or mean when the value of X is all of the observed values.

##	1	2	3	4	5	6	7	8
##	62.25973	61.44711	62.74731	62.80148	63.39740	59.98440	59.93022	60.68867
##	9	10	11	12	13	14	15	16
##	61.06789	60.95954	61.60964	62.25973	61.88051	62.63896	62.53061	62.58478
##	17	18	19	20	21	22	23	24
##	63.45158	63.66828	64.04750	64.91429	60.68867	60.74284	61.28459	60.95954
##	25	26	27	28	29	30	31	32
##	61.77216	61.55546	62.04303	62.04303	61.88051	62.25973	62.15138	62.63896
##	33	34	35	36	37	38	39	40
##	62.58478	65.78109	60.79702	61.23041	61.23041	60.79702	61.66381	61.55546
##	41	42	43	44	45	46	47	48
##	61.33876	61.71799	61.71799	61.39294	62.25973	61.88051	61.98886	62.04303
##	49	50	51	52	53	54	55	56
##	62.31391	62.25973	62.20556	62.04303	62.04303	62.09721	62.31391	62.31391
##	57	58	59	60	61	62	63	64
##	62.63896	62.90983	62.53061	62.69313	62.42226	62.74731	62.80148	62.85566
##	65	66	67	68	69	70	71	72
##	62.42226	62.74731	62.47643	63.12653	63.45158	63.01818	62.96400	63.28905
##	73	74	75	76	77	78	79	80
##	63.28905	63.23488	63.12653	63.12653	63.77663	63.88497	63.50575	64.10167
##	81	82	83	84	85	86	87	88
##	64.26420	64.58925	66.05196	60.36362	60.30944	60.68867	61.23041	61.28459
##	89	90	91	92	93	94	95	96
##	61.17624	60.90537	60.85119	61.55546	61.82634	61.71799	61.33876	61.77216
##	97	98	99	100	101	102	103	104
##	61.55546	61.33876	61.82634	61.50129	61.88051	62.20556	62.31391	62.09721
##	105	106	107	108	109	110	111	112
##	61.98886	62.20556	62.04303	62.09721	62.31391	62.36808	62.31391	61.98886
##	113	114	115	116	117	118	119	120
##	61.98886	62.25973	61.93469	62.69313	62.53061	62.53061	62.42226	62.58478
##	121	122	123	124	125	126	127	128
##	62.69313	62.63896	62.74731	62.80148	62.53061	62.74731	62.53061	62.90983
##	129	130	131	132	133	134	135	136
##	62.80148	62.42226	62.42226	62.85566	62.47643	62.90983	62.90983	63.07235
##	137	138	139	140	141	142	143	144
##	63.39740	63.07235	63.45158	63.28905	63.39740	63.28905	62.96400	63.23488
##	145	146	147	148	149	150	151	152

##	63.18070	63.45158	63.45158	63.01818	63.39740	63.39740	63.28905	63.61410
##	153	154	155	156	157	158	159	160
##	63.72245	63.61410	63.55993	63.66828	63.72245	63.55993	63.55993	63.93915
##	161	162	163	164	165	166	167	168
##	63.61410	63.55993	63.61410	63.50575	63.88497	63.77663	63.61410	63.83080
##	169	170	171	172	173	174	175	176
##	64.31837	64.21002	64.48090	64.37255	64.42672	64.15585	64.31837	64.04750
##	177	178	179	180	181	182	183	184
##	64.21002	64.15585	64.04750	64.04750	64.37255	64.15585	64.53507	64.53507
##	185	186	187	188	189	190	191	192
##	64.04750	64.53507	64.64342	65.07682	65.02264	65.29352	65.61856	65.83526
##	193	194	195	196	197	198	199	200
##	65.94361	67.13546	60.58032	61.06789	60.95954	61.66381	61.44711	61.39294
##	201	202	203	204	205	206	207	208
##	61.71799	61.77216	61.82634	61.82634	61.66381	62.36808	61.98886	62.31391
##	209	210	211	212	213	214	215	216
##	62.20556	61.88051	62.36808	61.88051	62.31391	62.36808	62.36808	61.88051
##	217	218	219	220	221	222	223	224
##	62.15138	62.25973	62.15138	62.04303	61.98886	62.04303	61.98886	62.42226
##	225	226	227	228	229	230	231	232
##	62.69313	62.80148	62.69313	62.47643	62.53061	62.85566	62.69313	62.90983
##	233	234	235	236	237	238	239	240
##	62.69313	62.90983	62.85566	62.90983	62.80148	62.69313	62.85566	62.53061
##	241	242	243	244	245	246	247	248
##	62.85566	63.18070	63.45158	63.34323	63.39740	63.34323	63.12653	62.96400
##	249	250	251	252	253	254	255	256
##	63.39740	63.34323	63.07235	63.28905	63.45158	63.45158	63.12653	63.28905
##	257	258	259	260	261	262	263	264
##	63.23488	63.39740	63.39740	62.96400	63.28905	63.12653	63.18070	63.12653
##	265	266	267	268	269	270	271	272
##	63.01818	63.34323	63.01818	63.12653	63.07235	63.12653	63.45158	63.34323
##	273	274	275	276	277	278	279	280
##	62.96400	63.01818	63.18070	63.72245	63.93915	63.77663	63.55993	63.93915
##	281	282	283	284	285	286	287	288
##	63.93915	63.55993	63.93915	63.55993	63.77663	63.93915	63.93915	63.66828
##	289	290	291	292	293	294	295	296
##	63.66828	63.88497	63.88497	63.99332	63.72245	63.83080	63.99332	63.77663
##	297	298	299	300	301	302	303	304
##	63.72245	63.55993	63.61410	63.72245	63.99332	63.55993	64.21002	64.53507
##	305	306	307	308	309	310	311	312
##	64.53507	64.53507	64.53507	64.37255	64.53507	64.31837	64.37255	64.42672
##	313	314	315	316	317	318	319	320
##	64.15585	64.37255	64.26420	64.53507	64.42672	65.07682	64.86012	64.58925
##	321	322	323	324	325	326	327	328
##	64.64342	64.64342	65.02264	64.75177	64.64342	64.86012	65.56439	65.51022
##	329	330	331	332	333	334	335	336
##	65.23934	65.18517	65.18517	65.61856	65.94361	65.99779	65.99779	61.01372
##	337	338	339	340	341	342	343	344
##	60.90537	61.17624	61.28459	61.50129	61.60964	61.77216	61.39294	61.77216
##	345	346	347	348	349	350	351	352
##	61.60964	61.71799	61.82634	61.60964	61.82634	61.44711	61.71799	62.36808
##	353	354	355	356	357	358	359	360
##	62.31391	62.25973	62.09721	61.98886	62.04303	61.88051	61.98886	62.25973
##	361	362	363	364	365	366	367	368
##	62.15138	62.20556	61.88051	62.09721	62.15138	62.04303	61.88051	62.31391
##	369	370	371	372	373	374	375	376
##	62.31391	62.09721	62.31391	62.58478	62.69313	62.74731	62.47643	62.53061
##	377	378	379	380	381	382	383	384
##	62.47643	62.53061	62.47643	62.80148	62.58478	62.90983	62.80148	62.85566
##	385	386	387	388	389	390	391	392
##	62.69313	62.47643	62.53061	62.63896	62.74731	62.63896	62.63896	62.47643

##	393	394	395	396	397	398	399	400
##	62.42226	62.69313	62.58478	62.74731	62.53061	62.69313	62.63896	62.80148
##	401	402	403	404	405	406	407	408
##	62.74731	62.47643	62.42226	62.42226	63.34323	63.39740	63.07235	63.28905
##	409	410	411	412	413	414	415	416
##	63.23488	63.01818	63.45158	63.07235	63.34323	63.28905	63.23488	63.07235
##	417	418	419	420	421	422	423	424
##	63.18070	63.01818	62.96400	63.12653	63.28905	62.96400	63.34323	63.39740
##	425	426	427	428	429	430	431	432
##	63.34323	63.12653	63.18070	63.01818	62.96400	63.12653	63.28905	63.45158
##	433	434	435	436	437	438	439	440
##	62.96400	63.01818	63.45158	63.01818	63.34323	63.12653	63.34323	63.50575
##	441	442	443	444	445	446	447	448
##	63.83080	63.55993	63.50575	63.77663	63.83080	63.61410	63.99332	63.77663
##	449	450	451	452	453	454	455	456
##	63.99332	63.61410	63.93915	63.61410	63.77663	63.72245	63.66828	63.55993
##	457	458	459	460	461	462	463	464
##	63.88497	63.77663	63.72245	63.72245	63.99332	63.61410	63.55993	63.99332
##	465	466	467	468	469	470	471	472
##	63.83080	63.83080	63.88497	63.55993	63.66828	63.55993	63.99332	63.83080
##	473	474	475	476	477	478	479	480
##	63.88497	63.88497	64.15585	64.42672	64.53507	64.04750	64.21002	64.53507
##	481	482	483	484	485	486	487	488
##	64.37255	64.42672	64.48090	64.10167	64.10167	64.21002	64.26420	64.04750
##	489	490	491	492	493	494	495	496
##	64.31837	64.42672	64.26420	64.37255	64.48090	64.31837	64.31837	64.15585
##	497	498	499	500	501	502	503	504
##	64.04750	64.31837	65.07682	64.80594	64.80594	64.75177	64.96847	64.86012
##	505	506	507	508	509	510	511	512
##	65.02264	64.58925	64.58925	64.96847	64.80594	64.58925	64.91429	65.23934
##	513	514	515	516	517	518	519	520
##	65.18517	65.18517	65.40187	65.56439	65.51022	65.45604	65.40187	65.45604
##	521	522	523	524	525	526	527	528
##	65.18517	65.72691	65.78109	66.10614	66.37701	60.63449	61.17624	61.12206
##	529	530	531	532	533	534	535	536
##	60.79702	60.85119	61.33876	61.44711	61.66381	61.39294	61.39294	61.50129
##	537	538	539	540	541	542	543	544
##	62.15138	62.04303	62.36808	61.88051	62.15138	61.93469	62.04303	62.25973
##	545	546	547	548	549	550	551	552
##	61.93469	62.04303	62.15138	61.98886	62.09721	61.88051	61.98886	62.09721
##	553	554	555	556	557	558	559	560
##	61.98886	62.31391	62.69313	62.80148	62.85566	62.53061	62.47643	62.53061
##	561	562	563	564	565	566	567	568
##	62.90983	62.69313	62.69313	62.53061	62.42226	62.47643	62.80148	62.47643
##	569	570	571	572	573	574	575	576
##	62.80148	62.80148	62.58478	62.58478	62.47643	62.85566	62.90983	62.63896
##	577	578	579	580	581	582	583	584
##	62.85566	62.47643	62.53061	62.47643	62.47643	62.90983	62.85566	63.45158
##	585	586	587	588	589	590	591	592
##	63.07235	63.01818	63.23488	63.28905	63.01818	63.12653	63.07235	63.18070
##	593	594	595	596	597	598	599	600
##	63.18070	63.28905	63.01818	63.45158	63.34323	62.96400	63.01818	63.39740
##	601	602	603	604	605	606	607	608
##	63.07235	63.01818	63.12653	63.01818	63.07235	63.45158	63.07235	63.45158
##	609	610	611	612	613	614	615	616
##	63.01818	63.07235	63.23488	63.12653	63.01818	63.12653	63.18070	63.45158
##	617	618	619	620	621	622	623	624
##	62.96400	63.23488	63.12653	63.18070	63.39740	63.39740	63.45158	63.28905
##	625	626	627	628	629	630	631	632
##	63.07235	63.88497	63.83080	63.66828	63.93915	63.55993	63.99332	63.55993
##	633	634	635	636	637	638	639	640

##	63.72245	63.77663	63.83080	63.50575	63.61410	63.50575	63.93915	63.93915
##	641	642	643	644	645	646	647	648
##	63.50575	63.50575	63.61410	63.93915	63.55993	63.50575	63.77663	63.61410
##	649	650	651	652	653	654	655	656
##	63.99332	63.88497	63.77663	63.88497	63.66828	63.50575	63.83080	63.88497
##	657	658	659	660	661	662	663	664
##	63.50575	63.66828	63.50575	63.77663	63.99332	63.50575	63.99332	63.50575
##	665	666	667	668	669	670	671	672
##	63.99332	63.61410	63.61410	63.50575	63.88497	63.55993	63.93915	64.31837
##	673	674	675	676	677	678	679	680
##	64.21002	64.31837	64.48090	64.15585	64.26420	64.37255	64.37255	64.42672
##	681	682	683	684	685	686	687	688
##	64.15585	64.37255	64.42672	64.42672	64.04750	64.48090	64.21002	64.53507
##	689	690	691	692	693	694	695	696
##	64.37255	64.10167	64.53507	64.48090	64.48090	64.15585	64.31837	64.37255
##	697	698	699	700	701	702	703	704
##	64.21002	64.42672	64.10167	64.10167	64.42672	64.26420	64.53507	64.21002
##	705	706	707	708	709	710	711	712
##	64.26420	64.64342	64.58925	65.02264	64.96847	64.69759	64.86012	64.96847
##	713	714	715	716	717	718	719	720
##	64.86012	64.96847	64.58925	65.07682	64.64342	65.07682	64.86012	64.58925
##	721	722	723	724	725	726	727	728
##	64.64342	64.75177	64.64342	65.40187	65.13099	65.23934	65.40187	65.13099
##	729	730	731	732	733	734	735	736
##	65.29352	65.45604	65.23934	65.13099	65.56439	66.05196	65.83526	65.88944
##	737	738	739	740	741	742	743	744
##	65.83526	61.12206	60.95954	61.39294	61.50129	61.82634	61.71799	62.04303
##	745	746	747	748	749	750	751	752
##	62.15138	62.31391	61.93469	62.04303	62.25973	62.80148	62.80148	62.90983
##	753	754	755	756	757	758	759	760
##	62.80148	62.63896	62.47643	62.53061	62.58478	62.80148	62.90983	62.63896
##	761	762	763	764	765	766	767	768
##	62.80148	62.69313	62.74731	62.42226	62.58478	62.63896	62.42226	62.90983
##	769	770	771	772	773	774	775	776
##	62.69313	62.80148	62.58478	62.80148	62.69313	63.28905	62.96400	63.34323
##	777	778	779	780	781	782	783	784
##	63.28905	63.28905	63.34323	63.28905	63.28905	63.39740	63.39740	63.12653
##	785	786	787	788	789	790	791	792
##	63.34323	63.07235	62.96400	63.39740	63.34323	63.07235	63.23488	62.96400
##	793	794	795	796	797	798	799	800
##	63.18070	62.96400	63.34323	62.96400	63.23488	63.12653	62.96400	63.01818
##	801	802	803	804	805	806	807	808
##	63.01818	63.39740	63.18070	63.23488	63.45158	63.83080	63.61410	63.99332
##	809	810	811	812	813	814	815	816
##	63.99332	63.83080	63.66828	63.99332	63.93915	63.83080	63.61410	63.93915
##	817	818	819	820	821	822	823	824
##	63.55993	63.83080	63.55993	63.83080	63.61410	63.61410	63.88497	63.93915
##	825	826	827	828	829	830	831	832
##	63.99332	63.50575	63.66828	63.83080	63.99332	63.99332	63.77663	63.55993
##	833	834	835	836	837	838	839	840
##	63.61410	63.83080	63.77663	63.72245	63.93915	63.88497	63.93915	63.66828
##	841	842	843	844	845	846	847	848
##	63.77663	63.77663	63.55993	64.15585	64.26420	64.26420	64.48090	64.53507
##	849	850	851	852	853	854	855	856
##	64.21002	64.31837	64.04750	64.53507	64.53507	64.48090	64.15585	64.48090
##	857	858	859	860	861	862	863	864
##	64.53507	64.21002	64.53507	64.15585	64.37255	64.37255	64.26420	64.42672
##	865	866	867	868	869	870	871	872
##	64.04750	64.31837	64.26420	64.10167	64.10167	64.15585	64.48090	64.15585
##	873	874	875	876	877	878	879	880
##	64.21002	64.42672	64.31837	64.42672	64.53507	64.42672	64.31837	64.26420

##	881	882	883	884	885	886	887	888
##	64.31837	64.53507	64.10167	64.86012	64.64342	64.64342	64.86012	64.96847
##	889	890	891	892	893	894	895	896
##	64.91429	65.02264	64.96847	64.58925	64.64342	64.64342	64.64342	64.69759
##	897	898	899	900	901	902	903	904
##	64.75177	64.64342	65.02264	64.96847	65.07682	65.02264	65.02264	64.69759
##	905	906	907	908	909	910	911	912
##	64.58925	65.02264	65.07682	64.58925	64.86012	64.75177	64.58925	65.61856
##	913	914	915	916	917	918	919	920
##	65.29352	65.51022	65.45604	65.13099	65.56439	65.51022	65.40187	65.40187
##	921	922	923	924	925	926	927	928
##	65.40187	65.56439	65.29352	65.23934	65.40187	65.34769	65.45604	65.18517
##	929	930	931	932	933	934	935	936
##	65.40187	65.34769	66.05196	65.67274	65.83526	65.88944	66.43119	66.64788
##	937	938	939	940	941	942	943	944
##	61.44711	62.20556	62.04303	62.15138	61.98886	61.88051	61.88051	62.42226
##	945	946	947	948	949	950	951	952
##	62.63896	62.80148	62.53061	62.85566	62.47643	62.63896	62.58478	62.63896
##	953	954	955	956	957	958	959	960
##	62.69313	63.34323	63.07235	63.18070	63.45158	63.23488	63.45158	63.18070
##	961	962	963	964	965	966	967	968
##	63.07235	63.12653	63.18070	63.39740	63.28905	63.01818	63.23488	63.01818
##	969	970	971	972	973	974	975	976
##	62.96400	62.96400	63.45158	63.93915	63.77663	63.77663	63.99332	63.88497
##	977	978	979	980	981	982	983	984
##	63.77663	63.88497	63.66828	63.66828	63.66828	63.99332	63.93915	63.61410
##	985	986	987	988	989	990	991	992
##	63.88497	63.77663	63.61410	63.77663	63.55993	63.66828	63.55993	63.72245
##	993	994	995	996	997	998	999	1000
##	63.61410	63.66828	63.88497	63.93915	63.50575	63.99332	63.66828	64.10167
##	1001	1002	1003	1004	1005	1006	1007	1008
##	64.48090	64.53507	64.21002	64.26420	64.26420	64.48090	64.26420	64.42672
##	1009	1010	1011	1012	1013	1014	1015	1016
##	64.26420	64.53507	64.15585	64.42672	64.48090	64.53507	64.21002	64.53507
##	1017	1018	1019	1020	1021	1022	1023	1024
##	64.31837	64.31837	64.10167	64.10167	64.48090	64.37255	64.15585	64.31837
##	1025	1026	1027	1028	1029	1030	1031	1032
##	64.21002	64.53507	64.53507	64.26420	64.31837	64.21002	64.15585	64.42672
##	1033	1034	1035	1036	1037	1038	1039	1040
##	64.04750	64.04750	64.64342	64.91429	64.96847	64.58925	64.69759	64.91429
##	1041	1042	1043	1044	1045	1046	1047	1048
##	64.91429	65.07682	64.91429	65.02264	64.64342	64.64342	64.96847	64.91429
##	1049	1050	1051	1052	1053	1054	1055	1056
##	65.07682	64.96847	65.02264	64.96847	65.02264	64.80594	65.02264	64.64342
##	1057	1058	1059	1060	1061	1062	1063	1064
##	64.75177	65.07682	64.58925	64.75177	65.45604	65.23934	65.23934	65.56439
##	1065	1066	1067	1068	1069	1070	1071	1072
##	65.56439	65.40187	65.51022	65.51022	65.23934	65.51022	65.18517	65.45604
##	1073	1074	1075	1076	1077	1078	1079	1080
##	65.45604	65.45604	65.13099	65.40187	65.78109	66.16031	65.67274	65.67274
##	1081	1082	1083	1084	1085	1086	1087	1088
##	65.67274	65.78109	65.88944	66.10614	66.16031	65.83526	65.99779	65.83526
##	1089	1090	1091	1092	1093	1094	1095	1096
##	66.59371	66.43119	66.70206	61.66381	62.04303	61.88051	62.31391	62.58478
##	1097	1098	1099	1100	1101	1102	1103	1104
##	62.90983	62.47643	62.58478	62.63896	62.80148	62.90983	62.90983	62.85566
##	1105	1106	1107	1108	1109	1110	1111	1112
##	62.74731	63.07235	62.96400	62.96400	63.39740	63.18070	63.34323	63.39740
##	1113	1114	1115	1116	1117	1118	1119	1120
##	63.28905	63.23488	62.96400	63.12653	63.45158	63.01818	63.34323	63.45158
##	1121	1122	1123	1124	1125	1126	1127	1128

##	63.88497	63.61410	63.50575	63.77663	63.93915	63.66828	63.61410	63.66828
##	1129	1130	1131	1132	1133	1134	1135	1136
##	63.55993	63.83080	63.55993	63.72245	63.66828	63.61410	63.99332	63.88497
##	1137	1138	1139	1140	1141	1142	1143	1144
##	63.55993	63.93915	63.88497	63.83080	63.61410	63.88497	63.99332	64.48090
##	1145	1146	1147	1148	1149	1150	1151	1152
##	64.10167	64.10167	64.42672	64.04750	64.31837	64.31837	64.48090	64.37255
##	1153	1154	1155	1156	1157	1158	1159	1160
##	64.48090	64.53507	64.26420	64.21002	64.10167	64.53507	64.42672	64.04750
##	1161	1162	1163	1164	1165	1166	1167	1168
##	64.15585	64.48090	64.31837	64.53507	64.21002	64.48090	64.26420	64.21002
##	1169	1170	1171	1172	1173	1174	1175	1176
##	64.31837	64.04750	64.31837	64.42672	65.02264	65.02264	65.02264	64.91429
##	1177	1178	1179	1180	1181	1182	1183	1184
##	64.64342	64.64342	64.75177	64.80594	64.75177	64.80594	65.02264	64.69759
##	1185	1186	1187	1188	1189	1190	1191	1192
##	64.91429	64.69759	64.91429	64.80594	64.91429	64.69759	64.75177	64.80594
##	1193	1194	1195	1196	1197	1198	1199	1200
##	65.07682	64.64342	65.51022	65.13099	65.56439	65.34769	65.61856	65.45604
##	1201	1202	1203	1204	1205	1206	1207	1208
##	65.18517	65.23934	65.61856	65.29352	65.18517	65.56439	65.18517	65.18517
##	1209	1210	1211	1212	1213	1214	1215	1216
##	65.23934	65.40187	66.10614	66.05196	65.72691	65.83526	66.05196	65.99779
##	1217	1218	1219	1220	1221	1222	1223	1224
##	66.05196	66.16031	65.99779	65.72691	65.78109	65.94361	65.83526	65.99779
##	1225	1226	1227	1228	1229	1230	1231	1232
##	66.10614	65.88944	66.37701	66.32284	66.70206	67.24381	66.86458	62.36808
##	1233	1234	1235	1236	1237	1238	1239	1240
##	61.88051	62.74731	62.63896	62.58478	62.74731	62.58478	62.85566	63.28905
##	1241	1242	1243	1244	1245	1246	1247	1248
##	63.18070	62.96400	63.07235	63.23488	63.23488	63.01818	63.28905	63.55993
##	1249	1250	1251	1252	1253	1254	1255	1256
##	63.83080	63.61410	63.72245	63.72245	63.99332	63.72245	63.50575	64.53507
##	1257	1258	1259	1260	1261	1262	1263	1264
##	64.37255	64.10167	64.48090	64.04750	64.42672	64.04750	64.48090	64.21002
##	1265	1266	1267	1268	1269	1270	1271	1272
##	64.42672	64.10167	64.15585	64.04750	64.10167	64.15585	64.69759	64.96847
##	1273	1274	1275	1276	1277	1278	1279	1280
##	64.86012	65.02264	65.07682	64.69759	64.91429	64.96847	64.69759	65.02264
##	1281	1282	1283	1284	1285	1286	1287	1288
##	64.69759	64.96847	64.64342	64.91429	65.45604	65.45604	65.61856	65.51022
##	1289	1290	1291	1292	1293	1294	1295	1296
##	65.51022	65.61856	65.18517	65.23934	65.13099	65.23934	65.13099	65.99779
##	1297	1298	1299	1300	1301	1302	1303	1304
##	66.05196	66.16031	65.94361	65.78109	66.05196	65.72691	65.88944	66.21449
##	1305	1306	1307	1308	1309	1310	1311	1312
##	66.26866	66.26866	66.48536	66.48536	67.13546	68.00225	63.45158	63.39740
##	1313	1314	1315	1316	1317	1318	1319	1320
##	63.66828	63.83080	63.88497	63.77663	63.88497	63.61410	64.15585	64.26420
##	1321	1322	1323	1324	1325	1326	1327	1328
##	64.15585	64.10167	64.42672	64.15585	64.80594	64.96847	65.02264	64.96847
##	1329	1330	1331	1332	1333	1334	1335	1336
##	65.02264	65.29352	65.13099	65.51022	65.56439	65.23934	65.45604	65.67274
##	1337	1338	1339	1340	1341	1342	1343	1344
##	66.05196	66.16031	66.21449	66.48536	66.32284	66.48536	66.64788	67.02711
##	1345	1346	1347	1348	1349	1350	1351	1352
##	67.73138	63.61410	64.10167	64.21002	64.58925	64.80594	64.69759	65.07682
##	1353	1354	1355	1356	1357	1358	1359	1360
##	64.80594	65.23934	65.61856	65.34769	65.78109	65.67274	65.88944	66.64788
##	1361	1362	1363	1364	1365	1366	1367	1368
##	67.02711	67.62303	63.77663	64.26420	65.07682	65.99779	66.32284	66.53953

```
##      1369      1370      1371      1372      1373      1374      1375
## 66.91876 67.56885 67.35216 65.13099 65.83526 68.27313 64.04750

# FITTED VALUES CONFIDENCE INTERVAL (CONFIDENCE INTERVAL FOR A CO
NDITIONAL MEAN is '63')
predict(m0, newdata=data.frame(mheight=c(63)), interval="confiden
ce", level=0.95)

##          fit          lwr          upr
## 1 64.0475 63.92441 64.17059

# SINGLE VALUE CONFIDENCE INTERVAL
predict(m0, newdata=data.frame(mheight=c(63)), interval="predictio
n")

##          fit          lwr          upr
## 1 64.0475 59.59999 68.49501

predict(m0, newdata=data.frame(mheight=c(63)), interval="predictio
n", level=0.95)

##          fit          lwr          upr
## 1 64.0475 59.59999 68.49501

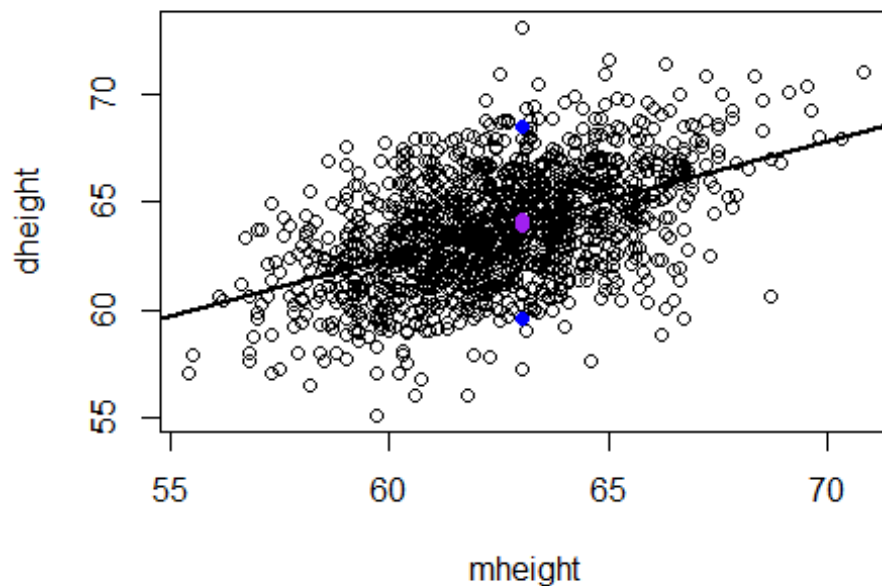
#### Plotting confidence intervals at x*=63

estimate <- predict(m0, newdata=data.frame(mheight=c(63)), interval="predictio
n")[1]

lowerpred <- predict(m0, newdata=data.frame(mheight=c(63)), interval="predicti
on")[2]
upperpred <- predict(m0, newdata=data.frame(mheight=c(63)), interval="predicti
on")[3]

upperfitted <- predict(m0, newdata=data.frame(mheight=c(63)), interval="confid
ence")[3]
lowerfitted <- predict(m0, newdata=data.frame(mheight=c(63)), interval="confid
ence")[2]

points(estimate~c(63), col="red", pch=19)
points(upperpred~c(63), col="blue", pch=19)
points(lowerpred~c(63), col="blue", pch=19)
points(upperfitted~c(63), col="purple", pch=19)
points(lowerfitted~c(63), col="purple", pch=19)
```



In the figure, we took the mean as '63' for example, and used the predict function to predict the fitted points and the prediction values and these functions' upper and lower limits(Blue dots).

```
#####
#PROBLEMS###
#####

#problem 2.2

names(UBSprices)

## [1] "bigmac2009" "bread2009" "rice2009" "bigmac2003" "bread2003"
## [6] "rice2003"

with(UBSprices, plot(rice2009~rice2003))

with(UBSprices, abline(lm(rice2009~rice2003), lwd=2))

with(UBSprices, abline(lm(log(rice2009)~log(rice2003))))
```

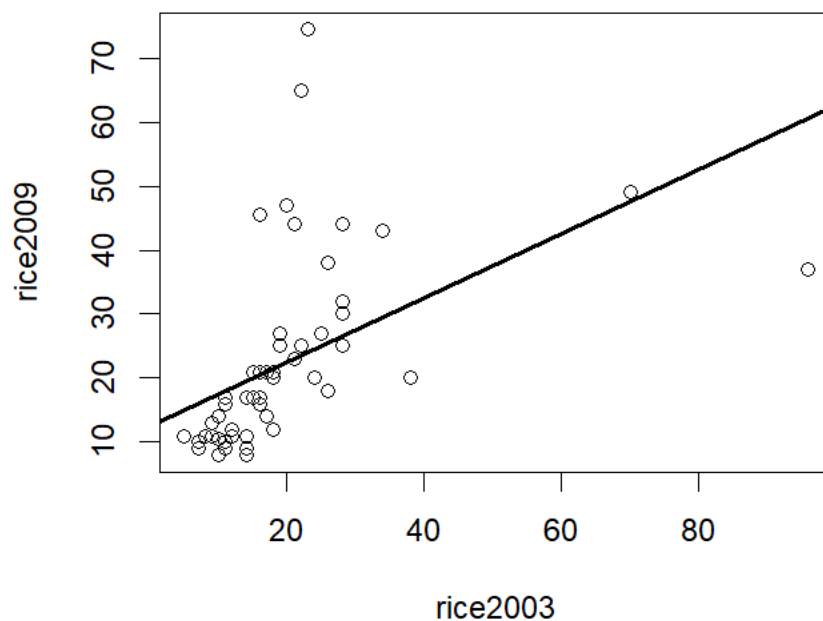
ANSWER-

(A) The points here on $Y=X$ are the cities where the price of rice in 2009 is equal to the price of rice in 2003. So, points that are above the mean are higher in price, and the line below the mean is lower in price compared to 2003 and 2009.

(B) when compared to the 2003 value of the country VILNIUS the price increased largely in 2009. Mumbai has the largest decreased prices compared from 2003 to 2009.

(C) from the figure, that the prices are not necessarily lower in 2009 than in 2003 as most of the points lie above the mean line have increased prices from 2003 to 2009 in those cities.

(D) fitting a simple linear regression in this problem is not appropriate because from 2003 to 2009 the values changed due to recession, we are not considering the values of the recession as the independent variable here so it makes a lot of errors to the present data.

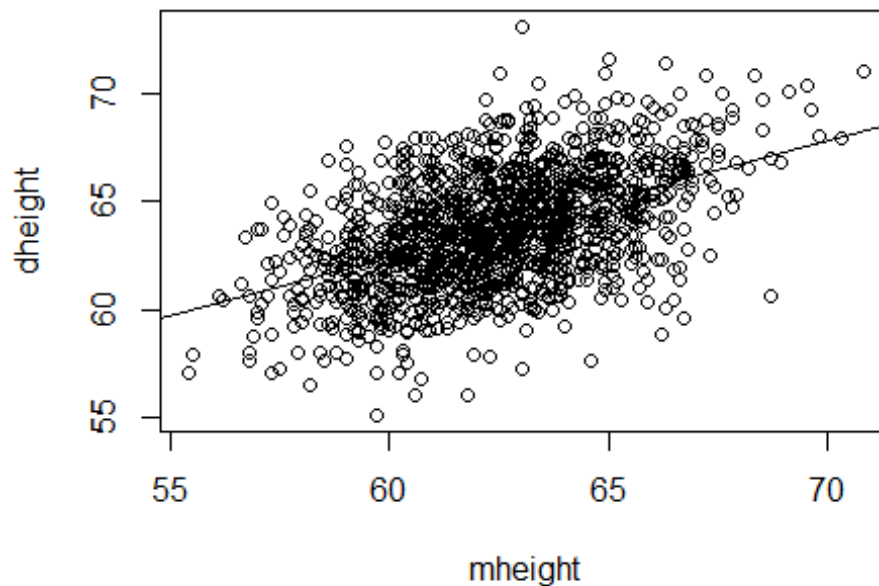


#problem 2.13(A)

```
names(Heights)

## [1] "mheight" "dheight"

with(Heights, plot(dheight~mheight))
with(Heights, abline(lm(dheight~mheight)))
```



```
with(Heights, summary(lm(dheight~mheight)))

##
## Call:
## lm(formula = dheight ~ mheight)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -7.397 -1.529  0.036  1.492  9.053
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  29.91744    1.62247   18.44  <2e-16 ***
## mheight       0.54175    0.02596   20.87  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 2.266 on 1373 degrees of freedom
```

```
## Multiple R-squared:  0.2408, Adjusted R-squared:  0.2402
## F-statistic: 435.5 on 1 and 1373 DF,  p-value: < 2.2e-16
```

ANSWER-

(A)

According to the summary,

Estimate of intercept - 29.91744

Std. Error intercept Estimate - 1.62247

Slope Estimate - 0.54175

Std. Error of slope Estimate - 0.02596

Multiple R-sqrd(coef of deter)- 0.2408

var(Heights)

```
##           mheight dheight
## mheight 5.546511 3.004806
## dheight 3.004806 6.760274
```

According to the mean equation, the slope is 0.54175, so a 1-inch increase in the mother's height results in a 0.54175 increase in the daughter's height. also, we have multiple R sqrd is 0.2408 which is a 24.08% variation in both daughters' and mothers' height.

(B)

$\alpha = 0.01$

99% Confidence Interval

Upper limit

$29.91744 + qt(1-0.01/2, 1373) * 0.54175$

```
## [1] 31.31484
```

Lower limit

$29.91744 - qt(1-0.01/2, 1373) * 0.54175$

```
## [1] 28.52004
```

confint(m0, level = 0.99)

```
##              0.5 %      99.5 %
## (Intercept) 25.7324151 34.1024585
## mheight      0.4747836 0.6087104
```

When the confidence interval is 99% then the slope value is 0.6087104.

#(C)99% of Prediction Interval

```
predict(m0, newdata=data.frame(mheight=c(64)),interval="prediction", level=0.99)
```

```
##      fit      lwr      upr
## 64.58925 58.74045 70.43805
```

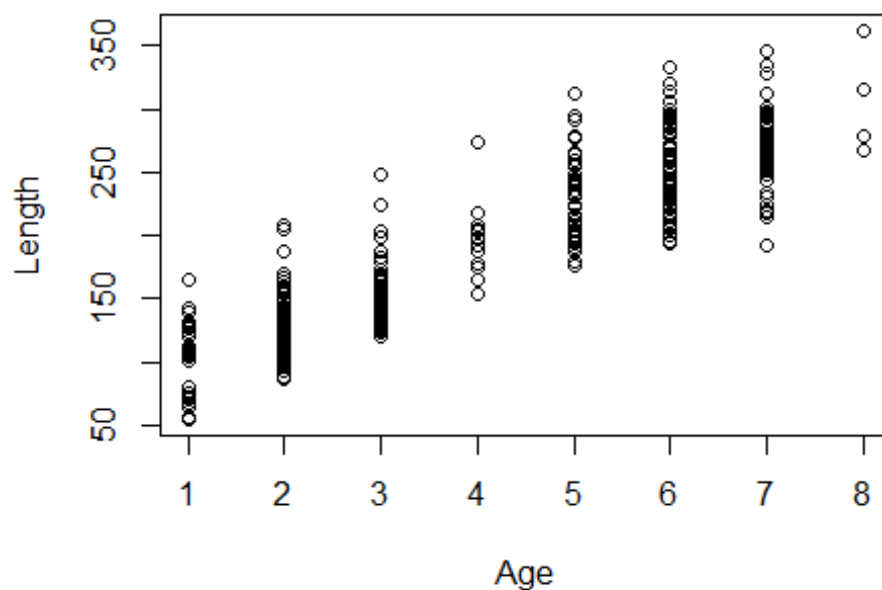
99% prediction interval for a daughter whose mother is 64 inches tall is between 70.43805 and 58.74045 inches and the fitted value is 64.58925 inches.

#Problem 2.15(A)

```
names(wblake)
```

```
## [1] "Age"    "Length" "Scale"
```

```
plot(Length~Age,wblake)
```




```
m1<-lm(Length~Age,wblake)
predict(m1, newdata=data.frame(Age=c(2,4,6)),interval="prediction", level=0.99)
```

ANSWER(A)

95% intervals for the mean length at ages 2, 4, and 6 years are:-

```
##      fit      lwr      upr
## 126.1749  51.87671 200.4732
## 186.8227 112.62705 261.0184
## 247.4705 173.20682 321.7342
```

#(B)

```
predict(m1, newdata=data.frame(Age=c(9)),interval="prediction", level=0.95)
```

The values are out of range there is no fish aged more than 8 we don't compute the mean function as there are no observations given.

```
##      fit      lwr      upr
## 1 338.4422 281.7056 395.1788
```

#Question 2.20(A)

```
plot(Interval~Duration,oldfaith)
abline(lm(Interval~Duration,oldfaith))
m2 <-lm(Interval~Duration,oldfaith)
predict(m2, newdata=data.frame(Duration=c(190)),interval="prediction")
```

```
##      fit      lwr      upr
## 1 67.59177 55.74792 79.43561
```

```
upperfitted <-predict(m2, newdata=data.frame(Duration=c(190)),interval="prediction")[2]
lowerfitted <-predict(m2, newdata=data.frame(Duration=c(190)),interval="prediction")[3]
points(upperfitted~c(190),col="purple",pch=19)
points(lowerfitted~c(190),col="purple",pch=19)
```

in the figure we did the regression function, for instance, we took a value of X as 190 sec then he have to wait for the next eruption between 55.74792 to 79.43561 minutes.

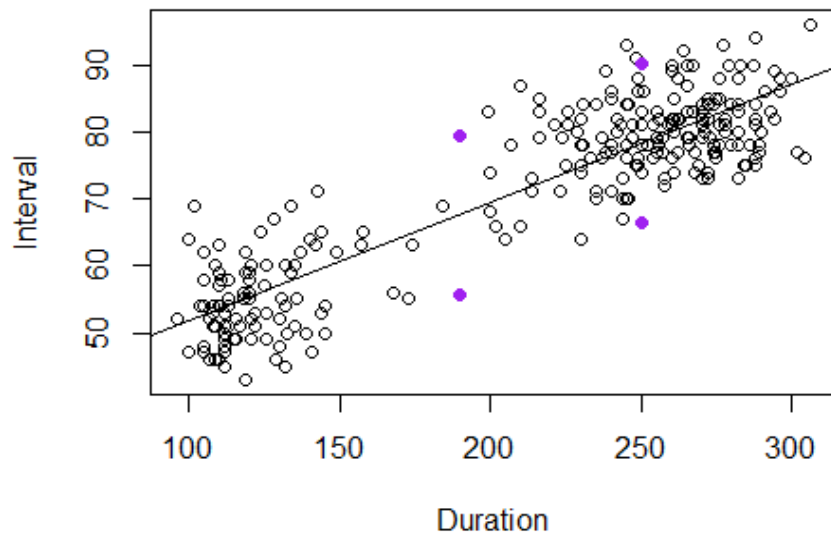
#(B)

```
predict(m2, newdata=data.frame(Duration=c(250)),interval="prediction", level = 0.95)
```

```
##      fit      lwr      upr
## 1 78.20354 66.35401 90.05307
```

```
upperfitted <-predict(m2, newdata=data.frame(Duration=c(250)),interval="prediction",level = 0.95)[2]
lowerfitted <-predict(m2, newdata=data.frame(Duration=c(250)),interval="prediction",level = 0.95)[3]
```

```
points(upperfitted~c(250),col="purple",pch=19)
points(lowerfitted~c(250),col="purple",pch=19)
```



Given that the eruption has lasted 250 sec then we have a 95% confidence interval for the time the individual will have to wait for the next eruption. So, according to the figure the individual has to wait between 66.35401-90.05307 minutes.

```
##(C) quantile of 90% the normal(0,1) distribution
qnorm(1-0.10/2)

## [1] 1.644854

predict(m2, newdata=data.frame(Duration=c(250)),interval="prediction", level
= 0.90)

##          fit          lwr          upr
## 1 78.20354 68.26967 88.13741
```